

POOL ARYANS

ALPHA

Algorithmic Long/Short Portfolio Holdings Allocation

Systemic Architecture and Strategy Framework for Alpha Extraction

A multi-strategy quantitative framework integrating **market microstructure, non-Gaussian statistical mechanics, machine learning, dynamic portfolio optimisation, and causal event-driven execution** across a twenty-stock equity universe.

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How to read the report

Data integrity builds a reliable market state. The six sleeves turn that state into alpha. The portfolio layer controls capital and risk. The exchange then converts the final target into realised PnL using the one bar delay and the competition constraints.

Data → Features → Alpha → Allocation → Risk → Execution → PnL

1 Systemic Architecture for Alpha Extraction

Financial markets are noisy, dynamic and structurally non-stationary. A trading signal that appears statistically attractive in isolation may cease to be profitable once transaction costs, position constraints, changing volatility regimes and interactions with other portfolio positions are incorporated.

The ALPHA problem is therefore not treated as a search for a single "best" indicator. We formulate it as a **constrained dynamic portfolio-allocation problem** in which multiple economically distinct alpha mechanisms compete for capital while operating under a common execution and risk-management framework.

The trading universe consists of twenty equities with an initial capital of

$$20 \times 100{,}000 = 2{,}000{,}000,$$

subject to a maximum of seven deployed strategies and a transaction cost of

$$c = 0.001$$

on the gross value of every purchase and sale.

At every point in time,

$$\sum_i \max(V_{i,t}, 0) \leq \text{Portfolio Value}_t,$$

and

$$\sum_i |\min(V_{i,t}, 0)| \leq \text{Portfolio Value}_t.$$

These constraints make capital allocation inseparable from signal generation.

1.1 Design Philosophy

Our architecture is built around four observations.

First, market behaviour is heterogeneous. Short-term dislocations may mean-revert, related assets may temporarily diverge, while persistent directional regimes demand trend-following rather than contrarian exposure.

Second, financial returns are non-Gaussian. Extreme observations occur far more frequently than implied by a normal distribution. Tail asymmetry and volatility clustering therefore require risk measures beyond conventional variance.

Third, alpha must survive implementation. A signal with small gross predictive power but excessive turnover can be economically worthless under the prescribed 0.1% fee.

Fourth, causality must be structural. Information observed during bar t may generate a target portfolio, but that target is executable only at the opening price of bar $t + 1$.

The system consequently separates three functions:

$$\boxed{\text{Alpha Generation} \rightarrow \text{Capital Allocation} \rightarrow \text{Execution}}$$

rather than embedding all three decisions inside individual strategies.

1.2 Strategy Structure

The portfolio does not allocate seven arbitrary fixed fractions of capital. Instead, each strategy produces a standardized target book corresponding to a different market phenomenon.

The relative-value layer captures temporary dislocations through gap mean reversion, cross sectional reversal and cointegrating spreads. The statistical-physics layer models asymmetric tail risk and changing persistence. The trend layer concentrates into sustained winners through breakout detection and pyramiding.

A machine-learning classifier is used separately as a *defensive risk overlay*. It forecasts conditions associated with severe overnight gaps and reduces exposure to the riskiest assets rather than opening speculative short positions.

The resulting strategy books are subsequently evaluated together by the portfolio allocator.

The architecture separates three fundamental questions:

Where does the alpha come from?
How much capital should it receive?
How can that capital be deployed without violating the rules?

2 Corporate Actions and Structural Data Integrity

The foundation of every subsequent statistical model is the continuity of the underlying price history. A mechanical corporate action that appears as an ordinary return can contaminate volatility, momentum, Hurst estimation, Tsallis tail fitting and machine-learning features simultaneously.

The framework therefore maintains **two different price representations**.

$$\boxed{\begin{aligned} &P_t^{\text{raw}} \rightarrow \text{Execution and realised P\&L}, \\ &P_t^{\text{adj}} \rightarrow \text{Features and statistical estimation}. \end{aligned}}$$

The execution engine continues to operate on the supplied raw market prices. Statistical calculations instead operate on a continuous analysis series in which mechanical events are neutralised.

2.1 Explicit Event Ledger

Large price discontinuities are screened and classified into one of three economically different categories.

For a detected split, define the observed price ratio

$$R_t = \frac{P_t}{P_{t-1}}.$$

If F_t denotes the corresponding correction factor, then the return used by the statistical layer is reconstructed as

$$r_t^{\text{adj}} = (1 + r_t^{\text{raw}})F_t - 1.$$

Thus a 2-for-1 or 3-for-1 stock split cannot be interpreted as a 50% or 67% market crash.

For structural distributions where the appropriate divisor cannot be reliably inferred from OHLCV data alone, the event is masked from rolling statistics rather than "corrected" using an arbitrary adjustment.

This is deliberately conservative: an unknown corporate distribution should contribute no artificial information rather than contaminated information.

2.2 Execution-Side Split Treatment

Corporate-action handling is also performed independently inside the execution engine.

If the price falls sufficiently to satisfy the competition split rule, the number of shares held is restated approximately as

$$N_t = \left\lceil N_{t-1} \frac{P_{t-1}}{P_t} \right\rceil .$$

The economic value of the position is therefore preserved through the mechanical price change.

The Buy and Hold benchmark is subjected to the same split-restatement logic. This prevents the strategy from appearing artificially superior merely because its own accounting correctly handles a corporate action while the benchmark absorbs a fictitious crash.

2.3 Feature Engineering

Once the price history is made statistically usable, the OHLCV stream is transformed into four principal families of features.

2.4 Range-Based Volatility

Traditional close-to-close volatility ignores the information contained inside the daily high–low range.

The Parkinson estimator exploits this additional information:

$$\sigma_{\mathrm{P}}^2 = \frac{1}{4N \ln 2} \sum_{t=1}^{N} \left[\ln \left(\frac{H_t}{L_t} \right) \right]^2 .$$

The Garman–Klass estimator additionally incorporates the opening and closing prices and is defined at the bar level by

$$\sigma_{\mathrm{GK}}^2 = \frac{1}{2} \left(\ln \frac{H_t}{L_t} \right)^2 - (2 \ln 2 - 1) \left(\ln \frac{C_t}{O_t} \right)^2 .$$

These measures feed both regime detection and position sizing.

Average True Range is separately used as the volatility unit in the breakout architecture.

2.5 Liquidity and Price Impact

The Amihud illiquidity measure approximates price impact using only OHLCV information:

$$ILLIQ_t = \frac{1}{N} \sum_{i=t-N+1}^{t} \frac{|r_i|}{P_i V_i}.$$

A security experiencing a large price response on small dollar volume is less attractive to a strategy whose statistical edge may only be tens of basis points.

Illiquidity is therefore used both as an exclusion criterion and as context for portfolio risk.

2.6 OHLCV-Based Order-Flow Proxy

True Order Flow Imbalance requires event-level limit-order-book information. Since only OHLCV data is available, the framework does not claim to observe true OFI.

Instead, it constructs a proxy combining

- Signed Volume
- Wick Imbalance
- Dollar-Volume Surprise

An extended lower wick on elevated volume, for example, indicates that aggressive selling was absorbed before the bar closed.

This does not reproduce the limit order book, but it provides a causal measure of observable supply–demand pressure using only permitted data.

2.7 Causal Information Flow

Every statistical feature used for trading is computed from a trailing window terminating no later than the current bar.

The complete information chain is

$$\boxed{\mathcal{I}_{\le t} \to X_t \to S_t \to W_t \to P_{t+1}^{\mathrm{open}}}$$

where X_t denotes features, S_t strategy signals and W_t the target portfolio.

No centred rolling statistic is required for signal construction, and a target formed on bar t cannot execute using bar t's closing price.

Data-layer principle.
The objective is not to create the largest possible collection of technical indicators. The objective is to transform OHLCV into a *causal representation of price behaviour, volatility, liquidity and market regime* without allowing accounting artefacts to masquerade as alpha.

3 Econophysics Foundations: Modelling Non-Gaussian Dynamics

Classical volatility tells us how large price moves are, but it does not fully show the shape of the tails or whether price changes are persistent. ALPHA therefore adds two related ideas: Tsallis statistics for tail shape and the Hurst exponent for behaviour through time.

3.1 The q Gaussian and Tail Asymmetry

The return density is represented by a q Gaussian,

$$p(x) = A_q \left[1 + (q-1)\beta(x-\mu)^2 \right]^{-1/(q-1)} .$$

As $q \to 1$, the model approaches a Gaussian. Larger q means heavier tails. Instead of estimating q from unstable sample kurtosis, the code uses the Student t relation

$$q = \frac{\nu + 3}{\nu + 1}, \qquad \nu = \frac{3 - q}{q - 1},$$

and estimates ν from rolling quantile ratios. This is more stable when extreme returns are present.

Losses and gains are fitted separately, producing (q_L, s_L) and (q_R, s_R). ALPHA then calculates Asymmetric Tsallis Relative Entropy, or ATRE, between the two fitted spectra. The full expression is

$$D_q(P_L \| P_R) = \frac{1}{q-1} \int P_L(x) \left[\left(\frac{P_L(x)}{P_R(x)} \right)^{q-1} - 1 \right] dx.$$

The interpretation is simple: ATRE is small when upside and downside returns have similar shapes and rises as their tail weight or scale becomes more different. This gives the system a direct measure of asymmetric tail risk.

3.2 Hurst and Multifractal State

Tail shape does not describe how returns are organised through time. For that, the engine uses the generalised Hurst exponent. For moment order m,

$$S_m(\tau) = \langle |X(t+\tau) - X(t)|^m \rangle \propto \tau^{mH(m)}.$$

The main reading is

$$H(2) > 0.5 \Rightarrow \text{persistent}, \qquad H(2) < 0.5 \Rightarrow \text{anti persistent}.$$

The code also monitors $H(1) - H(2)$ and autocorrelation in squared returns. A single Hurst value is therefore not used as a stand alone prediction; it is combined with volatility state before the regime sleeve trades.

These measurements feed directly into the strategy layer. The Tsallis sleeve uses the level of asymmetric tail risk, while the multifractal sleeve switches between following and fading price moves depending on the Hurst state.

> Volatility answers *how much is the market moving?* Tsallis adds *how different are the tails?* Hurst adds *are recent moves reinforcing or reversing?*

4 Alpha Generation I: Reversion & Relative Value

The first strategy family targets **temporary dislocations rather than persistent directional trends**. The common hypothesis is that short-lived liquidity shocks, cross sectional overextension, and deviations from stationary inter-asset relationships can create prices that temporarily depart from their local equilibrium.

Importantly, the three strategies define equilibrium differently. Gap Reversion operates through time within a single asset; Cross-Sectional Reversal compares assets at the same point in time; and GARCH Pairs Trading models the relative equilibrium between two assets.

4.1 Strategy 1: Gap-Down Mean Reversion

A large overnight decline can represent either genuine information-driven repricing or a temporary liquidity concession. Blindly buying every gap-down therefore mixes two economically different populations.

ALPHA attempts to isolate **exhaustion-driven gaps** by requiring three simultaneous conditions.

First, the overnight gap must be unusually negative relative to the asset's own recent gap distribution:

$$z_t^{\text{gap}} = \frac{g_t - \mu_{g,t}}{\sigma_{g,t}} \leq z_{\text{entry}}. \tag{1}$$

In the submitted configuration,

$$z_{\text{entry}} = -1.3102. \tag{2}$$

Second, the preceding session must exhibit range expansion relative to its recent baseline. Third, trading volume must be abnormally elevated:

$$z_t^V \geq 2. \tag{3}$$

These filters are evaluated using only information available before the trade decision. Their role is to separate high-participation exhaustion events from ordinary downward drift or genuine repricing.

Once triggered, the strategy takes a long position and scales exposure inversely to realised volatility:

$$w_{i,t} \propto \frac{I_{i,t}^{\text{gap}}}{\sigma_{i,t}}, \tag{4}$$

where $I_{i,t}^{\text{gap}}$ denotes the active gap-reversion state.

The trade is held only for a finite horizon and is subject to a hard loss threshold. A failed mean reversion trade is therefore not allowed to become an indefinite directional position.

4.2 Strategy 2: Cross-Sectional Reversal

The second strategy asks a different question:

Which assets are unusually overextended relative to the rest of the universe at the current point in time?

Each asset receives a volume-conditioned, volatility-scaled momentum score. Assets are then ranked cross sectionally rather than against a fixed absolute threshold.

The strategy structure is

$$\text{Long} \quad \leftarrow \quad \text{Low Momentum Rank}, \tag{5}$$

$$\text{Short} \quad \leftarrow \quad \text{High Momentum Rank}. \tag{6}$$

The resulting portfolio is explicitly dollar neutral:

$$\sum_i w_{i,t} = 0. \tag{7}$$

This suppresses much of the common market direction and makes strategy returns depend primarily on the relative performance of recent losers and winners.

Within each leg, the portfolio additionally incorporates a volatility premium tilt:

$$\widetilde{w}_{i,t} = w_{i,t} \left(\sigma_{i,t}^{P}\right)^{\gamma}, \tag{8}$$

where $\sigma_{i,t}^{P}$ denotes Parkinson volatility and the submitted configuration uses

$$\gamma = 0.6254. \tag{9}$$

Volatility is not rewarded indiscriminately. Assets in the most illiquid portion of the cross-section, as measured by the Amihud ranking, are excluded before the long and short legs are constructed.

The objective is therefore to capture relative overextension and a cross sectional volatility premium without allowing poor liquidity to consume the expected edge through implementation friction.

4.3 Strategy 3: GARCH Cointegration Pairs

Correlation alone does not imply that the distance between two asset prices should converge. The pairs strategy therefore requires a stronger property: **stationarity of the relative pricing relationship**.

For each candidate pair A, B, the formation window estimates

$$\log P_{A,t} = \alpha + \beta \log P_{B,t} + \varepsilon_t. \tag{10}$$

The resulting spread is

$$S_t = \log P_{A,t} - \beta \log P_{B,t} - \alpha. \tag{11}$$

An Augmented Dickey–Fuller test is applied to S_t. Only candidate relationships rejecting the unit-root null at the configured significance level are retained.

The hedge ratio is then **frozen for the subsequent trading block**. This prevents the current divergence from redefining its own equilibrium.

A fixed spread standard deviation would implicitly assume constant risk. ALPHA instead models the conditional spread variance using a GARCH$(1,1)$ process:

$$h_t = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G h_{t-1}. \tag{12}$$

The standardized divergence becomes

$$z_t = \frac{S_t}{\sqrt{h_t}}. \tag{13}$$

The strategy is governed by a simple state machine:

$$|z_t| > 2 \quad \Rightarrow \quad \text{Enter against divergence}, \tag{14}$$

$$|z_t| < 0.5 \quad \Rightarrow \quad \text{Exit}, \tag{15}$$

$$|z_t| > 4 \quad \Rightarrow \quad \text{Stop / Relationship Failure}. \tag{16}$$

Thus, the same absolute spread deviation is interpreted differently in calm and turbulent spread regimes.

The strategy remains market neutral by constructing equal-and-opposite long and short exposure across the selected pair.

Common principle, different equilibrium definitions.
The three strategies share a belief in temporary disequilibrium, but deliberately disagree on how equilibrium should be defined:

an asset's own history, the cross-section, or a stationary inter-asset relationship.

This reduces dependence on any single form of mean reversion.

5 Alpha Generation II: Tail Risk, Trend & Regime

The previous strategy family assumes some form of equilibrium toward which prices can revert. That assumption becomes dangerous during persistent directional moves or structural changes in the return distribution.

The second alpha family is therefore designed for precisely those conditions in which conventional mean reversion logic becomes unreliable.

5.1 Strategy 4: Tsallis Asymmetric Tail-Risk Premium

The Tsallis sleeve trades a property of the **return distribution itself**, rather than simply the path followed by prices.

An important distinction from the initial research hypothesis is that the final implementation does not trade changes in the left-tail q-parameter as a direct arbitrage signal. Empirical testing instead supported the **level of asymmetric tail risk** as the more useful cross sectional signal.

The interpretation is therefore:

$$\begin{aligned} &\text{Higher Asymmetric Tail Risk} \\ \Longrightarrow\ &\text{Higher Required Expected Return.} \end{aligned} \tag{17}$$

The signal combines the cross sectional ranks of smoothed Asymmetric Tsallis Relative Entropy and downside tail shape:

$$Score_{i,t} = \frac{Rank(ATRE_{i,t}) + Rank(q_{L,i,t})}{2}. \tag{18}$$

The strategy takes

$$\text{Long} \quad \leftarrow \quad \text{High Tail-Risk Score}, \tag{19}$$

$$\text{Short} \quad \leftarrow \quad \text{Low Tail-Risk Score}. \tag{20}$$

The fitted tail statistics are exponentially smoothed before trading. This is deliberate: asymmetric tail structure evolves more slowly than ordinary bar-to-bar price noise, and trading every fluctuation would generate unnecessary turnover under the prescribed transaction cost.

The submitted implementation therefore treats ATRE primarily as a **risk-premium signal rather than a short horizon reversal signal**.

A co-skewness throttle provides an additional risk-control layer. Positions whose downside behaviour is unusually aligned with extreme market states receive reduced weight, preventing the sleeve from mistaking correlated crash exposure for diversification.

5.2 Strategy 5: Donchian Breakout with Pyramiding

Most strategies in the architecture continuously rebalance positions toward a target. This creates an important structural weakness: successful positions are repeatedly trimmed as they appreciate.

Buy and Hold behaves differently. A sufficiently successful asset naturally becomes a larger fraction of the portfolio.

To compete with this effect without relying on leverage, ALPHA requires a mechanism capable of progressively concentrating into sustained winners. The Donchian breakout sleeve provides that mechanism.

A long breakout occurs when the current adjusted price exceeds the highest price of the preceding N bars:

$$P_t > \max\left(P_{t-1}, P_{t-2}, \ldots, P_{t-N}\right). \tag{21}$$

The short-side condition is defined symmetrically:

$$P_t < \min\left(P_{t-1}, P_{t-2}, \ldots, P_{t-N}\right). \tag{22}$$

The Donchian channel is shifted by one bar, so the breakout bar itself can never be included in the level against which it is compared.

Position sizing is based on Wilder's Average True Range. Define the relative volatility unit

$$N_t = \frac{ATR_t}{P_t}. \tag{23}$$

The approximate weight of one risk unit is then

$$w_t^{\text{unit}} = \frac{f_{\text{risk}}}{k_{\text{stop}} N_t}, \tag{24}$$

where f_{risk} represents the amount of sleeve capital risked by one unit and k_{stop} controls stop distance.

Thus,

$$\text{Higher Volatility} \quad \Rightarrow \quad \text{Smaller Position}. \tag{25}$$

The defining feature of this strategy is **pyramiding**. After a successful entry, an additional unit is added whenever the price advances sufficiently in the direction of the trade:

$$\Delta P \geq 0.5\, ATR. \tag{26}$$

This process continues up to the configured maximum number of units.

The portfolio therefore behaves asymmetrically:

$$\boxed{\text{Cut Failed Breakouts} \quad \text{and} \quad \text{Concentrate into Persistent Winners}} \tag{27}$$

The submitted implementation permits up to six units and applies a separate scaling factor to the short leg. This reflects the different economic role of long and short trend exposure within a portfolio whose overall objective remains positively exposed to the equity premium.

5.3 Strategy 6: Multifractal Regime Switching

The final alpha sleeve asks a different question:

Is the market currently persistent, anti-persistent, or approximately random?

Using the generalised Hurst exponent introduced previously,

$$H(2) > H_{\text{trend}} \tag{28}$$

indicates a persistent regime, while

$$H(2) < H_{\text{revert}} \tag{29}$$

indicates an anti-persistent regime.

The submitted configuration uses

$$H_{\text{trend}} = 0.6029, \qquad H_{\text{revert}} = 0.5000. \tag{30}$$

A Hurst reading alone is not treated as sufficient evidence of a tradeable regime. The scaling state must occur simultaneously with a volatility surge:

$$\frac{\sigma_{GK,t}}{\sigma_{GK,t}^{\text{slow}}} \geq 1.2. \tag{31}$$

This conjunction reduces the probability of treating a noisy Hurst estimate in an inactive market as a genuine structural transition.

Direction then depends on the regime.

For a persistent state,

$$H(2) \geq H_{\text{trend}} \quad \Rightarrow \quad \text{Follow the Prevailing Move}. \tag{32}$$

For an anti-persistent state,

$$H(2) \leq H_{\text{revert}} \quad \Rightarrow \quad \text{Fade the Prevailing Move}. \tag{33}$$

The Hurst exponent is therefore used as a **two-sided regime classifier**, rather than merely as a trend filter.

The exit mechanism is an adaptive trailing stop. Its width depends on the persistence of squared returns.

When volatility clustering remains strong, the position is given greater room to develop. As the autocorrelation of squared returns decays, the stop tightens, allowing the strategy to exit as the regime loses statistical persistence.

Why this family is necessary.
The first alpha family profits when equilibrium is restored.
This second family exists for the opposite possibility:

equilibrium itself may be changing.

Tail-risk repricing, persistent breakouts and changing multifractal structure therefore provide complementary return sources when mean reversion assumptions are weakest.

6 Dynamic Portfolio Construction & Risk Management

Generating six target books is only the first step. The strategies can disagree on direction, trade at different speeds and carry different kinds of risk. The portfolio layer therefore decides how much capital each sleeve receives before any order reaches the exchange.

6.1 LightGBM Gap Risk Overlay

Machine learning is used as a defensive layer, not as a directional return model. LightGBM estimates

$$p_{i,t} = P(\text{severe gap}_{i,t+1} \mid \mathcal{I}_{\le t}).$$

Inputs include range expansion, volume and dollar volume surprise, scaled momentum, volatility state, Amihud rank, Tsallis variables, wick geometry, the OHLCV flow proxy and recent reversal. The model uses a 500 bar training window, refits every 63 bars and leaves a 5 bar embargo before the next scored block. Predicted risk is ranked across the universe. The highest risk names can be scaled down to 50% of their intended exposure. In other words, the model answers *where should we hold less?*

6.2 Cost Aware Mean CVaR Allocation

The allocator sees sleeve returns after an estimate of their own turnover cost,

$$r^{net}_{k,t} = r^{gross}_{k,t} - 0.001\, TO_{k,t}.$$

Capital is then distributed with a Mean CVaR objective,

$$\min_{\mathbf{a}} \left[CVaR_{0.95}(\mathbf{a}) - \lambda \mu^{\top}\mathbf{a} + \gamma \|\mathbf{a} - \mathbf{a}_{prev}\|_1 \right],$$

subject to $\sum_k a_k = 1$ and sleeve concentration limits. CVaR focuses on the bad tail of realised sleeve scenarios, while the turnover term discourages large changes in allocation. Ledoit Wolf shrinkage is used only for the risk based prior or fallback where covariance is actually needed; the CVaR linear program itself works directly with return scenarios.

6.3 Capital Deployment and Final Risk Controls

A portfolio made mostly of neutral or event driven strategies can leave capital unused. The framework therefore adds a long inverse volatility core,

$$w^{core}_{i,t} \propto \frac{1}{\sigma_{i,t}},$$

which provides broad long exposure without replacing the alpha sleeves. The combined book then passes through a per name cap, signal smoothing, weight smoothing, trailing stop quarantine, cooldown, a no trade band and independent long and short exposure projection.

Sleeves → Net Cost Scenarios → Mean CVaR → LightGBM → Long Core → Risk Projection.

7 Event-Driven Backtesting & Execution Integrity

A strategy can look profitable and still be invalid if it uses a price that was not available when the signal was formed. The backtester therefore keeps cash, shares, fees, corporate actions and exposure inside the simulation.

7.1 One Bar Execution Delay

Every target follows the same timing rule,

$$\boxed{\text{Close}(t) \rightarrow \text{Signal}(t) \rightarrow \text{Target}(t) \rightarrow \text{Open}(t+1)}.$$

The target generated after bar t is stored as pending. At the next open, the old pending target is executed first and only then is the portfolio marked at the next close. The delay is therefore structural rather than a return series shift added after the fact.

7.2 Share and Cash Accounting

Portfolio equity is

$$E_t = C_t + \sum_i n_{i,t} P_{i,t},$$

where negative shares represent shorts. Desired market values are converted to whole shares at the actual execution price. Sales are processed before purchases so released cash can fund new positions in the same rebalance.

Every buy and sell pays

$$Fee = 0.001 |\Delta n| P.$$

Purchases are bounded by available cash. A no trade band skips small changes unless the trade reduces risk or the total book has drifted far enough that the portfolio needs to catch up with its target.

7.3 Exposure and Corporate Actions

Long and short exposure are measured separately,

$$L_t = \frac{\sum_i \max(n_i P_i, 0)}{E_t}, \qquad S_t = \frac{\sum_i |\min(n_i P_i, 0)|}{E_t}.$$

Targets are projected before execution and realised exposure is checked again at open and close. If market movement pushes a side too close to the rulebook limit, the engine reduces that side rather than simply recording a breach.

Corporate actions are handled separately from the adjusted feature series. When the competition split condition is

triggered while a position is open, share count is restated so the economic value of the position is preserved. The passive benchmark receives the same treatment.

The system uses a fixed random seed and no wall clock dependent trading logic, so the same input and configuration are designed to reproduce the same path.

The backtest asks what the account could actually own, pay for and execute using information available at that point in time.

8 Performance Evaluation

The performance section uses the realised share level equity curve after the one bar delay, whole share rounding, fees and exposure checks. The passive benchmark starts with the same capital and follows the same split convention.

The main comparison is

$$R_{excess} = R_{ALPHA} - R_{Buy\&Hold}.$$

Sharpe is calculated from realised daily returns,

$$SR = \frac{\bar{r}}{\sigma_r}\sqrt{252},$$

and drawdown is measured from the running equity peak,

$$DD_t = \frac{E_t}{\max_{\tau \le t} E_\tau} - 1.$$

The final report keeps only the metrics that explain return, risk and trading cost: Total Return, Net PnL, Sharpe, Sortino, Maximum Drawdown, number of trades, win rate, total fees and average turnover. The large calendar year, strategy allocation and extended PnL tables are omitted from the main report; they remain available in the code output if the judges ask for them.

Final run values: Return 283.48%, Buy and Hold 127.07%, Sharpe 1.076, Sortino 1.421, Max Drawdown -31.06%, Net PnL 5,669,532, Fees 227,921, Trades 20,568, Win Rate 57.02%.

8.1 Equity Curve and Interpretation

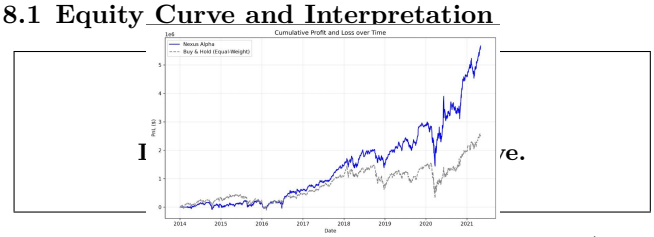


Table 1: Headline performance vs. Buy and Hold (realised run).

Metric	ALPHA	Buy & Hold	Excess
Total Return	283.48%	127.07%	156.41%
Sharpe Ratio	1.076	0.676	0.400
Max Drawdown	-31.06%	-34.58%	3.52%

The curve is kept because terminal return alone can hide whether performance is steady or dominated by one short interval. A stronger final return is useful only when it is read together with Sharpe, drawdown and the fee bill.

Signals → Allocation → Open$(t+1)$ → Fees → Constraints → Realised PnL.

9 Robustness, Generalisation & Ablation

Strong historical performance is not proof of a durable edge. The framework therefore keeps every estimation step behind the period it later trades.

9.1 Walk Forward Separation

LightGBM trains only on earlier observations and uses an embargo before the next scored block. Cointegration, hedge ratios and GARCH parameters are estimated on a formation window and then frozen while the next block trades. Mean CVaR allocation uses only sleeve returns already realised before the current decision.

Past Data → Estimate → Embargo → Unseen Block → Roll.

This setup does not prove that a future regime will behave like the past. It only prevents the test period from teaching the model how to trade itself.

9.2 What Was Removed

An earlier short horizon OFI scalp was removed because turnover consumed the edge after fees. The first Tsallis design, which traded changes in q, was also rejected; the final version trades the level of asymmetric tail risk instead. Removing a plausible idea when it fails realistic validation is part of the research process, not a weakness of it.

9.3 Leave One Out Attribution

Standalone strategy return does not tell us whether a sleeve helps the final portfolio. Leave one out testing removes a sleeve before Mean CVaR allocation and lets the remaining sleeves use the freed capital. The comparison then asks whether Total Return, Sharpe and Maximum Drawdown improve or weaken. The full ablation table is omitted here to save space and can be attached after the final run if required.

Robustness here means that adaptive steps are learned from earlier data and used on later data. It does not mean future profitability is guaranteed.

10 Limitations, Compliance & Conclusion

The framework is broad but it has clear limits. OHLCV data cannot reveal queue position, exact bid ask spread or true order book imbalance, so flow variables are proxies. Cointegrating relationships can break. Hurst and tail parameters are finite window estimates and can be noisy. The backtester includes fees, delayed execution, whole shares, cash and exposure limits, but it cannot reproduce endogenous market impact. Complex corporate actions with no reliable divisor are removed from statistical memory rather than adjusted using a guess.

10.1 Compliance with the ALPHA Mandate

The implementation starts with INR 2,000,000 across 20 assets. It uses six alpha generating sleeves plus a capital

deployment core. LightGBM is a defensive risk overlay rather than another directional alpha sleeve. Every buy and sell pays 0.1%. A decision formed at bar t can first trade at Open(t+1). Positions are stored as whole shares, purchases use available cash, and long and short exposure are checked independently against current portfolio value. Corporate actions are treated in both the statistical and execution layers, and the passive benchmark follows the same split rule.

10.2 Operational Checks

Rule	Implementation
Capital	INR 2,000,000 across 20 assets
Strategy count	6 alpha sleeves + capital deployment core
Machine learning	Defensive LightGBM exposure scaler
Transaction fee	0.1% on every buy and sell
Execution timing	Signal at t, first fill at Open($t+1$)
Positions	Whole shares with explicit cash accounting
Exposure	Long and short sides checked independently
Corporate actions	Separate feature and execution treatment

10.3 What the Framework Contributes

The framework adds value in six places: it separates raw execution prices from clean analysis prices; models both ordinary volatility and asymmetric tails; combines reversion, relative value and trend sleeves; uses machine learning only where its role is clear; allocates capital using net-of-cost strategy history; and treats execution rules as part of the strategy rather than as a final reporting adjustment.

10.4 Conclusion

ALPHA is built around one main idea:

A trading signal has value only if it survives portfolio construction, risk control and realistic execution.

The process starts with data integrity, builds a market state, creates six different alpha views, allocates capital using net-of-cost history, projects the desired book into the feasible region and finally lets the exchange create realised PnL. The strategies are not six unrelated backtests. They are parts of one system in which signal, capital and execution remain separate.

The final causal chain is

$$\mathcal{I}_{\leq t} \rightarrow \text{Decision}_t \rightarrow \text{Execution at Open}_{t+1} \rightarrow \text{Realised PnL}.$$

ALPHA is not one prediction model. It is a complete decision system.

Market State → Alpha → Allocation → Risk → Execution → PnL

11 Appendix: Performance Tables

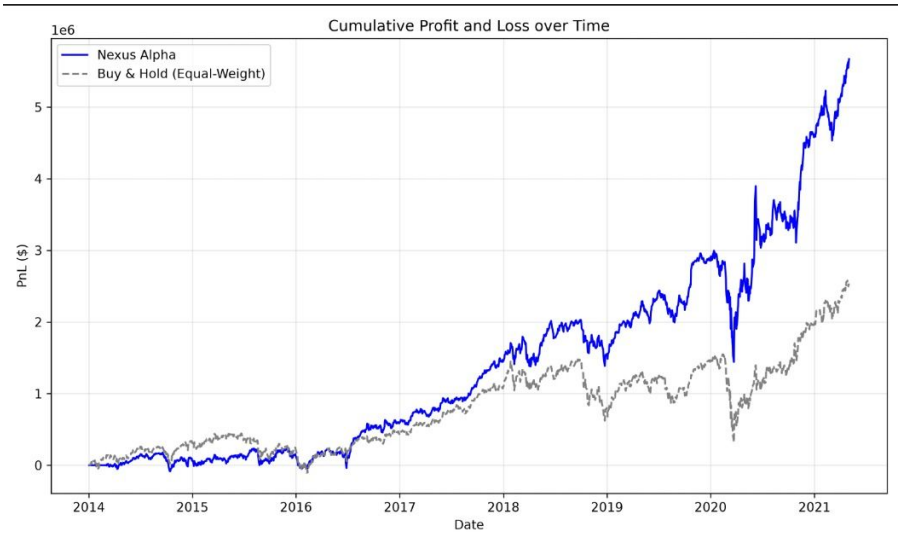
The tables below summarise realised performance at the asset level and at the strategy-sleeve level, alongside the passive Buy and Hold benchmark. All figures are realised, share-level results from the final run, measured on the basis described in Section 8.

11.1 Asset-Wise Performance vs. Buy and Hold

Per-asset realised metrics for the ALPHA book compared against a passive Buy and Hold position in the same instrument, both measured after fees and the one-bar execution delay.

Asset	Strategy Return	B&H Return	Strategy Sharpe	B&H Sharpe	Strategy Max DD	B&H Max DD	Excess Return
Asset 01	4.74%	136.32%	0.54	0.58	-2.21%	-45.24%	-131.58%
Asset 02	16.39%	-14.43%	0.39	0.21	-9.38%	-84.53%	30.81%
Asset 03	4.42%	113.39%	0.29	0.50	-4.30%	-48.83%	-108.97%
Asset 04	17.67%	212.53%	0.53	0.78	-9.05%	-31.79%	-194.86%
Asset 05	12.94%	-29.53%	0.91	0.13	-2.59%	-78.30%	42.47%
Asset 06	0.90%	50.44%	0.09	0.35	-4.23%	-45.46%	-49.54%
Asset 07	-0.23%	-17.19%	0.01	0.11	-11.74%	-72.43%	16.96%
Asset 08	12.83%	133.63%	0.59	0.51	-4.74%	-49.53%	-120.80%
Asset 09	22.62%	950.67%	0.80	0.95	-6.74%	-64.95%	-928.05%
Asset 10	11.85%	212.46%	0.67	0.73	-3.87%	-43.63%	-200.61%
Asset 11	1.86%	32.15%	0.09	0.31	-9.44%	-65.71%	-30.30%
Asset 12	17.32%	-76.04%	0.61	-0.17	-8.10%	-92.47%	93.36%
Asset 13	-6.50%	-24.08%	-0.19	0.13	-9.56%	-56.90%	17.58%
Asset 14	3.95%	-8.27%	0.34	0.14	-2.90%	-68.18%	12.22%
Asset 15	0.16%	25.07%	0.32	0.27	-0.15%	-58.10%	-24.90%
Asset 16	3.93%	5.05%	0.35	0.18	-2.09%	-67.44%	-1.13%
Asset 17	5.62%	-2.37%	0.46	0.20	-3.39%	-54.52%	7.99%
Asset 18	0.04%	45.14%	0.01	0.32	-7.03%	-52.92%	-45.10%
Asset 19	4.56%	-11.01%	0.26	0.12	-5.42%	-70.68%	15.57%
Asset 20	5.11%	-6.87%	0.42	0.12	-2.71%	-55.22%	11.97%

Note: Asset 01 to Asset 20 follow the ordering of the supplied twenty-stock universe. Every column is computed from the realised share-level equity curve, after fees and after the one-bar execution delay.



sleeve (Section 6.1); its effect is reflected inside each sleeve's net performance.